

PHYS 121 — HW3 Solutions

WRITTER

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Q1. Gravitation from a ball (8 pts)

Shell theorem: A uniform spherical shell of matter exerts no gravitational force on a particle inside it, and exerts the same gravitational force on a particle outside it as if all the shell's mass were concentrated at its center.

(2) Gravitational field $g(r)$: For a uniform solid sphere of radius R and mass M :

- **Outside ($r \geq R$):** The sphere acts as a point mass at the center, so

$$g(r) = \frac{GM}{r^2}, \quad r \geq R.$$

- **Inside ($r < R$):** Only the mass inside radius r contributes. The mass inside is $M(r) = M(r/R)^3$, so

$$g(r) = \frac{GM(r)}{r^2} = \frac{GM}{R^3}r, \quad r < R.$$

(3) Gravitational potential energy $U(r)$: With $U(\infty) = 0$:

- **Outside ($r \geq R$):**

$$U(r) = -\frac{GMm}{r}, \quad r \geq R.$$

- **Inside ($r < R$):** Integrate from infinity, ensuring continuity at $r = R$:

$$U(r) = U(R) + \int_R^r g(r')m \, dr' = -\frac{GMm}{R} + \frac{GMm}{2R^3}(r^2 - R^2),$$

$$U(r) = -\frac{GMm}{2R} \left(3 - \frac{r^2}{R^2} \right), \quad r < R.$$

At $r = R$, both expressions give $U(R) = -GMm/R$, ensuring continuity.

(4) Evaluations:

$$|U(R/2)| = \left| -\frac{GMm}{2R} \left(3 - \frac{1}{4} \right) \right| = \frac{11GMm}{8R},$$

$$|U(R)| = \frac{GMm}{R},$$

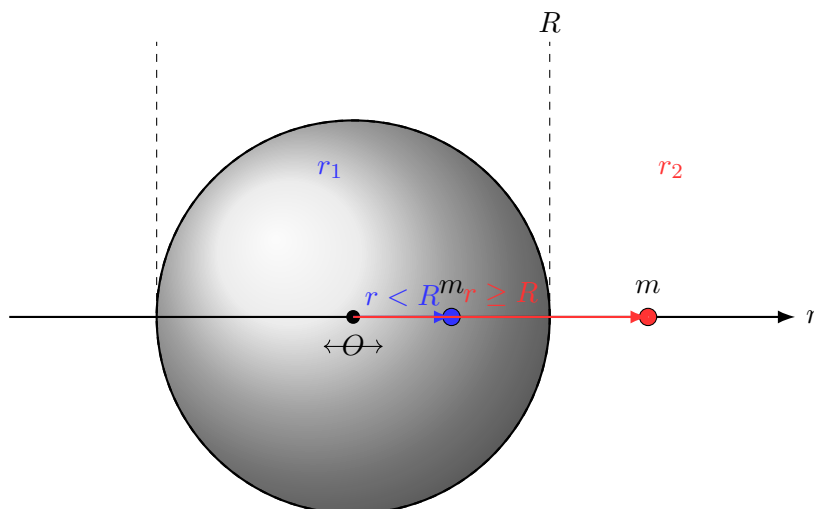
$$|U(2R)| = \frac{GMm}{2R}.$$

Ranking: $|U(R/2)| > |U(R)| > |U(2R)|$. The magnitude is largest at $r = R/2$ because the test mass is deeper in the potential well.

(i) Inside the sphere, $g(r) = (GM/R^3)r$ varies *linearly* with r .

(ii) As $r \rightarrow \infty$, $U(r) = -GMm/r \rightarrow 0^-$ (approaches zero from below).

(1) Drawing:



Q2. Explore Vesta (10 pts)

Given: diameter $D = 520$ km, so radius $R = 260$ km $= 2.60 \times 10^5$ m, mass $M = 2.67 \times 10^{20}$ kg.

(1) **Escape speed:**

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2 \times 6.674 \times 10^{-11} \times 2.67 \times 10^{20}}{2.60 \times 10^5}} = \boxed{370 \text{ m/s}}.$$

(2) **Orbital period:** For a circular orbit at altitude $h = 15$ km above the surface, the orbital radius is $r = R + h = 2.75 \times 10^5$ m. Using Kepler's third law:

$$\begin{aligned} T &= 2\pi \sqrt{\frac{r^3}{GM}} = 2\pi \sqrt{\frac{(2.75 \times 10^5)^3}{6.674 \times 10^{-11} \times 2.67 \times 10^{20}}} \\ &= 2\pi \sqrt{\frac{2.08 \times 10^{16}}{1.78 \times 10^{10}}} = 2\pi \times 1080 = \boxed{6.79 \times 10^3 \text{ s} = 1.89 \text{ h}}. \end{aligned}$$

(3) **Reflection:** A spherical model is only marginally useful because Vesta is not spherical—it's an irregular asteroid with significant deviations from sphericity. Real orbits are tricky due to:

- Non-uniform mass distribution (density variations)
- Irregular shape causing non-central gravitational field
- Rotational effects and potential tumbling
- Surface features (craters, ridges) that create gravitational anomalies

These factors cause orbital perturbations, precession, and instability that a simple spherical model cannot capture.

Q3. Kepler's law—Pluto's small moons (8 pts)

Given: Charon orbits at $r_C = 19\,600$ km with period $T_C = 6.39$ d. Two small satellites at $r_1 = 48\,000$ km and $r_2 = 64\,000$ km.

Kepler's third law: $T^2 \propto r^3$, so $T \propto r^{3/2}$. Scaling from Charon:

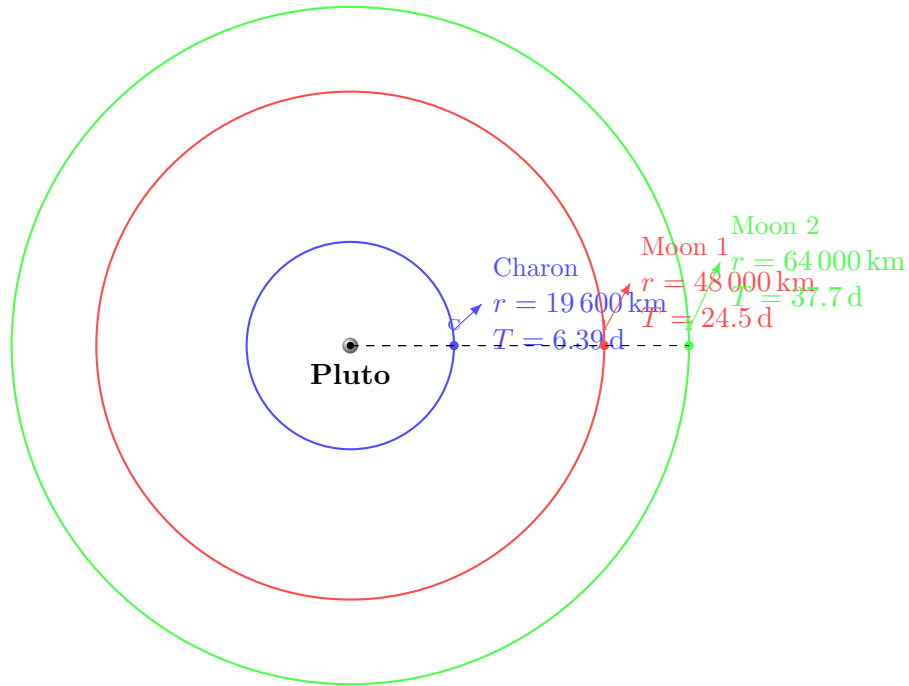
$$\frac{T}{T_C} = \left(\frac{r}{r_C}\right)^{3/2} \Rightarrow T = T_C \left(\frac{r}{r_C}\right)^{3/2}.$$

For $r_1 = 48\,000$ km:

$$T_1 = 6.39 \times \left(\frac{48000}{19600} \right)^{3/2} = 6.39 \times (2.449)^{3/2} = 6.39 \times 3.83 = \boxed{24.5 \text{ d} = 588 \text{ h}}.$$

For $r_2 = 64\,000$ km:

$$T_2 = 6.39 \times \left(\frac{64000}{19600} \right)^{3/2} = 6.39 \times (3.265)^{3/2} = 6.39 \times 5.90 = \boxed{37.7 \text{ d} = 905 \text{ h}}.$$



Q4. Gravity and SHM in a straight tunnel (14 pts)

For a uniform spherical planet of radius R and density ρ , the mass inside radius r is $M(r) = \frac{4}{3}\pi r^3 \rho$.

(2) Gravitational force: At distance r from the center ($r \leq R$), only the mass inside r contributes:

$$F(r) = -\frac{GM(r)m}{r^2} = -\frac{G \cdot \frac{4}{3}\pi r^3 \rho \cdot m}{r^2} = -\frac{4\pi G \rho m}{3} r.$$

The force is proportional to r and directed toward the center (restoring).

(3) Simple harmonic motion: Newton's second law gives

$$m \frac{d^2 r}{dt^2} = -\frac{4\pi G \rho m}{3} r \quad \Rightarrow \quad \frac{d^2 r}{dt^2} = -\frac{4\pi G \rho}{3} r.$$

This is SHM with angular frequency

$$\omega = \sqrt{\frac{4\pi G \rho}{3}}.$$

(4) Period:

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{3}{4\pi G \rho}} = \sqrt{\frac{3\pi}{G \rho}}.$$

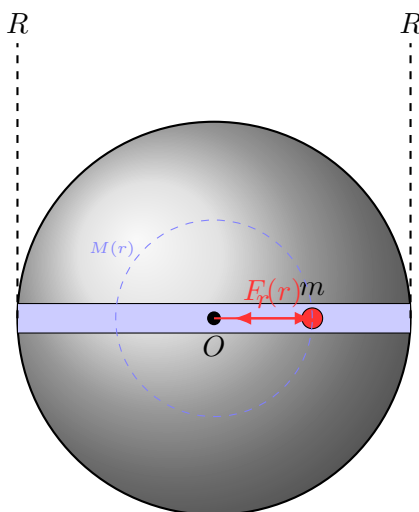
(5) Comparison with circular orbit: For a circular orbit skimming the surface ($r = R$), the orbital speed is $v = \sqrt{GM/R}$ where $M = \frac{4}{3}\pi R^3 \rho$. The period is

$$T_{\text{orbit}} = \frac{2\pi R}{v} = 2\pi R \sqrt{\frac{R}{GM}} = 2\pi \sqrt{\frac{R^3}{G \cdot \frac{4}{3}\pi R^3 \rho}} = \sqrt{\frac{3\pi}{G\rho}}.$$

Thus $T = T_{\text{orbit}}$: the tunnel period equals the surface-skimming orbital period.

(6) Check: If density doubles ($\rho \rightarrow 2\rho$), then $T = \sqrt{3\pi/(G\rho)} \propto 1/\sqrt{\rho}$, so the period decreases by a factor of $\sqrt{2}$.

(1) Drawing:



Q5. Pendulum in an elevator (10 pts)

For a pendulum of length ℓ , the period is $T = 2\pi\sqrt{\ell/g_{\text{eff}}}$, where g_{eff} is the effective gravitational acceleration in the elevator's reference frame.

(1) Cases:

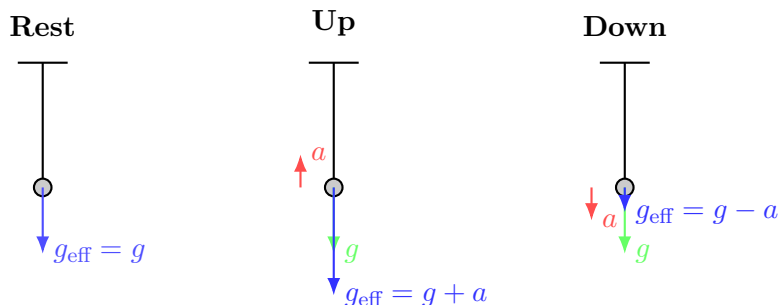
- **Rest:** $g_{\text{eff}} = g$, so $T_0 = 2\pi\sqrt{\ell/g}$.
- **Accelerating upward at a :** In the elevator frame, effective gravity is $g_{\text{eff}} = g + a$, so $T_{\text{up}} = 2\pi\sqrt{\ell/(g + a)}$.
- **Accelerating downward at a (with $a < g$):** Effective gravity is $g_{\text{eff}} = g - a$, so $T_{\text{down}} = 2\pi\sqrt{\ell/(g - a)}$.

(3) Numerical values: With $\ell = 0.90$ m and $a = 2.0$ m/s²:

$$\begin{aligned} T_0 &= 2\pi\sqrt{\frac{0.90}{9.8}} = \boxed{1.90 \text{ s}}, \\ T_{\text{up}} &= 2\pi\sqrt{\frac{0.90}{9.8 + 2.0}} = 2\pi\sqrt{\frac{0.90}{11.8}} = \boxed{1.74 \text{ s}}, \\ T_{\text{down}} &= 2\pi\sqrt{\frac{0.90}{9.8 - 2.0}} = 2\pi\sqrt{\frac{0.90}{7.8}} = \boxed{2.13 \text{ s}}. \end{aligned}$$

Ranking: $T_{\text{down}} > T_0 > T_{\text{up}}$. The period is longest when accelerating downward (smallest effective g).

(4) **Free fall:** In free fall, $a = g$, so $g_{\text{eff}} = g - g = 0$. The period becomes infinite—the pendulum doesn't oscillate; it floats relative to the elevator.



Q6. Van der Waals $\approx$ Hooke near the minimum (8 pts)

Given potential: $U(r) = U_0 \left[\left(\frac{R_0}{r} \right)^{12} - 2 \left(\frac{R_0}{r} \right)^6 \right]$.

(1) **Sketch:** The minimum occurs when $dU/dr = 0$. At $r = R_0$, $U(R_0) = U_0(1 - 2) = -U_0$.

(2) **Conservative force:**

$$F(r) = -\frac{dU}{dr} = -U_0 \left[-12 \frac{R_0^{12}}{r^{13}} + 12 \frac{R_0^6}{r^7} \right] = 12U_0 R_0^6 \left(\frac{R_0^6}{r^{13}} - \frac{1}{r^7} \right).$$

(3) **Expansion near minimum:** Let $r = R_0 + r'$ with $|r'| \ll R_0$. For small r' :

$$\frac{R_0}{r} = \frac{R_0}{R_0 + r'} = \frac{1}{1 + r'/R_0} \approx 1 - \frac{r'}{R_0} + \dots$$

More precisely, expand $F(r)$:

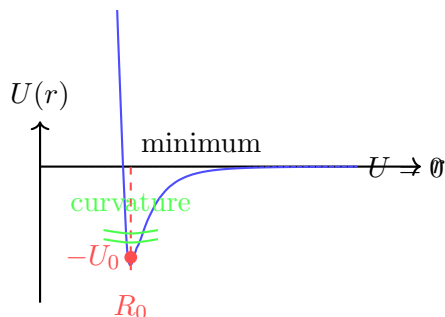
$$F(R_0 + r') = 12U_0 R_0^6 \left[\frac{R_0^6}{(R_0 + r')^{13}} - \frac{1}{(R_0 + r')^7} \right].$$

Using $(R_0 + r')^{-n} \approx R_0^{-n}(1 - nr'/R_0)$ for small r' :

$$\begin{aligned} F(R_0 + r') &\approx 12U_0 R_0^6 \left[\frac{1}{R_0^7} \left(1 - \frac{13r'}{R_0} \right) - \frac{1}{R_0^7} \left(1 - \frac{7r'}{R_0} \right) \right] \\ &= 12U_0 R_0^6 \cdot \frac{1}{R_0^7} \left(-\frac{13r'}{R_0} + \frac{7r'}{R_0} \right) \\ &= -\frac{72U_0}{R_0^2} r'. \end{aligned}$$

Thus $F \approx -kr'$ with $k = \frac{72U_0}{R_0^2}$.

(4) **Check:** Units: $[k] = [U_0]/[R_0^2] = \text{J/m}^2 = \text{N/m}$ (correct for spring constant). The force is restoring: $F = -kr'$ means F opposes displacement from equilibrium.



Q7. A moving pulse (9 pts)

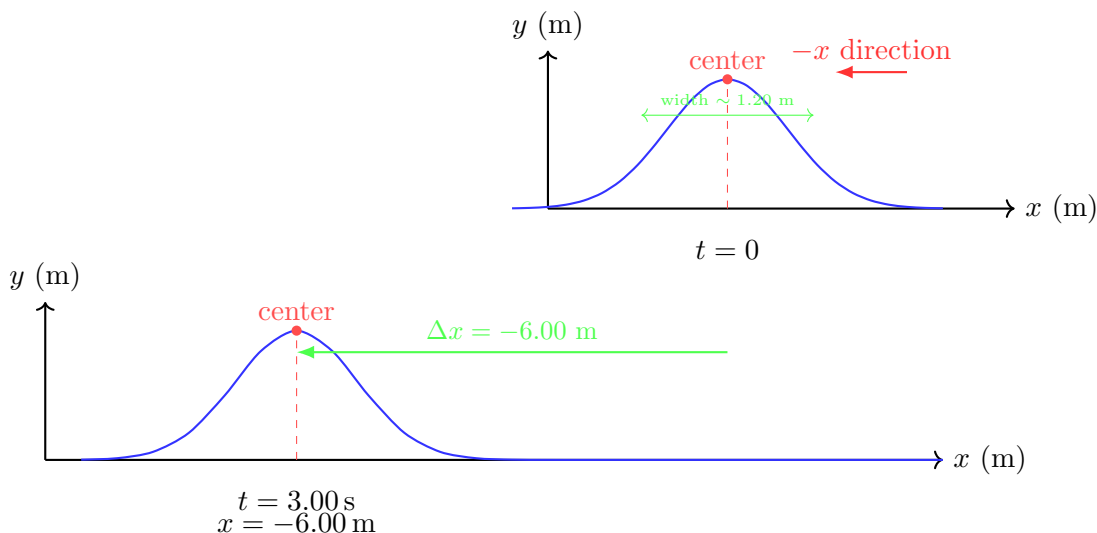
Given: $y(x, t) = 4.20 \text{ m} \cdot e^{-(x+2.00t)^2/(1.20)^2}$.

(1) At $t = 0$: $y(x, 0) = 4.20 \text{ m} \cdot e^{-(x/1.20)^2}$. This is a Gaussian pulse centered at $x = 0$ with width $\sim 1.20 \text{ m}$.

(2) Rewrite: $y(x, t) = 4.20 \text{ m} \cdot e^{-(x+2.00t)^2/(1.20)^2} = f(x + 2.00t)$ where $f(u) = 4.20 \text{ m} \cdot e^{-u^2/(1.20)^2}$.

The form $f(x + vt)$ indicates motion in the $-x$ **direction** with speed $v = 2.00 \text{ m/s}$.

(3) Center movement: In 3.00 s , the center moves $\Delta x = -v\Delta t = -2.00 \times 3.00 = -6.00 \text{ m}$ (to the left).



Q8. Wave on a string (10 pts)

Given: $y(x, t) = A \cos(kx - \omega t + \varphi) = 0.15 \text{ m} \cos(0.15 \text{ m}^{-1} \cdot x + 1.50 \text{ s}^{-1} \cdot t + 0.25)$.

Note: The signs are $+kx$ and $+\omega t$, so we rewrite as $y = A \cos(kx + \omega t + \varphi)$.

(1) Wave speed and direction:

$$v = \frac{\omega}{k} = \frac{1.50 \text{ s}^{-1}}{0.15 \text{ m}^{-1}} = 10.0 \text{ m/s}.$$

The form $\cos(kx + \omega t)$ indicates motion in the $-x$ **direction** (leftward).

(2) At $x = 0.40 \text{ m}$, $t = 5.00 \text{ s}$:

$$\begin{aligned} kx + \omega t + \varphi &= 0.15 \times 0.40 + 1.50 \times 5.00 + 0.25 \\ &= 0.06 + 7.50 + 0.25 = 7.810. \end{aligned}$$

$$y = 0.15 \text{ m} \cos(7.810) = \boxed{0.007 \text{ m}},$$

$$\begin{aligned} v_y &= \frac{\partial y}{\partial t} = -A\omega \sin(kx + \omega t + \varphi) \\ &= -0.15 \text{ m} \times 1.50 \text{ s}^{-1} \sin(7.810) = \boxed{-0.225 \text{ m/s}}, \end{aligned}$$

$$\begin{aligned} a_y &= \frac{\partial^2 y}{\partial t^2} = -A\omega^2 \cos(kx + \omega t + \varphi) \\ &= -0.15 \text{ m} \times (1.50 \text{ s}^{-1})^2 \cos(7.810) = \boxed{-0.015 \text{ m/s}^2}. \end{aligned}$$

Bonus. Wave in different mediums (0–2 pts)

When you shout near a pond, sound waves in air (a compressional wave with speed $\sim 343 \text{ m/s}$) encounter the water surface. At the air–water interface, most sound energy is *reflected* because of the large impedance mismatch (water’s density and bulk modulus are much higher than air’s). However, a small fraction *transmits* into water, where sound travels faster ($\sim 1500 \text{ m/s}$) due to water’s higher density and bulk modulus. The transmitted wave has a different wavelength ($\lambda_{\text{water}} = v_{\text{water}}/f$) but the same frequency as the incident wave. A detector in water would register this transmitted component, though it would be much weaker than the original sound in air due to the reflection loss at the interface.

Assistance notes:

LLMs were used as translators and concept explainers. No direct solutions were provided.